# Kasra Jamshidi

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## Education

**PhD Computer Science** @ Simon Fraser University 2019-2024

Thesis: *Designing Application-Aware Systems for Mining Large Graph Data*, advised by Prof. Keval Vora.

**BSc Hon Computer Science** @ Simon Fraser University 2014-2019

Thesis: *Asynchronous Graph Mining*, advised by Prof. Keval Vora.

## Publications

**Contigra: Graph Mining with Containment Constraints** EuroSys '24

Joanna Che, **Kasra Jamshidi**, Keval Vora.
European Conference on Computer Systems, April 2024.

**Scalable Byzantine Fault Tolerant Analytics without Replication** PPoPP '24

**Kasra Jamshidi**, Keval Vora.
Symposium on Principles and Practice of Parallel Programming, March 2024.

**Accelerating Graph Mining Systems with Subgraph Morphing** EuroSys '23

**Kasra Jamshidi**, Guoqing Harry Xu, Keval Vora.
European Conference on Computer Systems, May 2023.

**Anti-Vertex For Neighborhood Constraints In Subgraph Queries** GRADES-NDA '22

**Kasra Jamshidi**, Mugilan Mariappan, Keval Vora.
ACM Workshop on Graph Data Management Experiences & Systems and Network Data Analytics, June 2022.

**A Deeper Dive Into Pattern-Aware Subgraph Exploration With Peregrine** OSR '21

**Kasra Jamshidi**, Keval Vora.
SIGOPS Operating Systems Review 55, 1, June 2021.

**Peregrine: A Pattern-Aware Graph Mining System** EuroSys '20

**Kasra Jamshidi**, Rakesh Mahadasa, Keval Vora.
European Conference on Computer Systems, April 2020.

## Work Experience

**Software Developer** @ Safe Software November 2023 - Present

- Designed and implemented integrations with 10+ RDBMS for the flagship product FME, a data automation engine written in C++ and deployed to 200K active users in 25K organizations including Google, Amazon, and Oracle
- Implemented batching optimization for DuckDB integration, allowing FME to leverage DuckDB's columnar data layout for **100x improvement in processing throughput** over standard row-based processing
- Planned and led Neo4j graph database integration using Python, designing efficient interfaces between graph data and FME's internal C++ data representation, fulfilling multiple user-submitted feature requests for graph support
- Planned and led Delta Lake integration, enabling efficient access to delta tables in cloud storage
- Acted as domain expert on Python development within team and on graph databases throughout the organization
- Proactively improved code clarity by building C++23 library polyfills and a compile-time enum handling library during company hackathon, eliminating ad-hoc, redundant, and untested utility methods throughout the codebase
- In the year since joining, the team achieved **2x higher task** throughput than projected by management and the team's maintenance backlog became empty for the first time in 17 years

### **Graduate Research Assistant** @ SFU PDCL — April 2019 - September 2024

- Designed and implemented Peregrine, a programmable parallel graph mining system that is **700x faster** than the previous state-of-the-art with **8x fewer CPUs**, while using **100x less memory.**
  - Maintain open-source project: https://github.com/pdclab/peregrine.
  - Performance scales nearly ideally with physical CPU cores (e.g., 48 cores lead to 41x speedup).
  - Custom lockfree aggregator.

- Built a distributed, fault tolerant stream processing system for an RDMA-enabled cluster using C++23. Solves analytics queries on massive, rapidly updating data, sustaining an average output throughput of **200M (3.5GB) records per second**.
  - Custom lockfree arena allocator to reduce context switches in critical path.
  - Custom Paxos implementation to take advantage of RDMA and provide Byzantine fault tolerance.
  - Asynchronous RDMA network layer implementation.

- Developed a runtime-agnostic query optimization framework that automatically improves graph mining execution speed by **10-34x** (saving **24 hours+** on some queries) with overhead in the milliseconds.
  - Accounts for low-level runtime traits to fix multiple different bottlenecks, uncovered via extensive profiling.
  - Formally proven correct with arbitrary aggregations and practically scales to large patterns and data graphs.
  - Integrated and evaluated the framework in 4 existing graph mining systems.

### **Undergraduate Research Assistant** @ SFU PDCL — September 2018 - August 2019

- Developed a distributed graph mining model without the synchronization requirements of Arabesque (SOSP '15) and implemented a proof-of-concept using Java, Scala, and the Akka actor framework.
- Implemented the DualSIM (SIGMOD '16) disk-based pattern-matching algorithm in C++.

### **Founding Developer** @ Polly Language Exchange/Lingvu — January 2017 - March 2018

- Developed web chat app that pairs users seeking to learn each other's native languages for learning opportunities.
- Technologies: WebRTC, Angular2, NGINX, Lua, Redis, Phoenix/Elixir, PostgreSQL Geospatial, Vagrant.

### **Software Intern** @ Nexedi Inc. — June 2016 - January 2017

- Developed various React web applications, including implementing reverse-indexing and fuzzy full-text search.
- Wrote technical documentation and tutorials for new products.
- Assisted CEO with technical workshops teaching prospective customers about products.

## Teaching & Service

### **Stanford Code In Place Volunteer Instructor** — 2025

Volunteer section lead for Stanford's free online introductory programming course, based on material from CS106A. Designed and delivered 60 minute interactive tutorials to a class of 12 beginner programmers over 6 weeks.

### **Teaching Assistantships** — 2020-2023

Fall 2020 - CMPT 479 | Fall 2022 - CMPT 431 | Spring 2023 - CMPT 431 | Fall 2023 - CMPT 431, CMPT 479.

### **Student Mentoring** — 2020-2024

- Joanna Che (MSc), *Graph Mining with Containment Constraints.*
- Rakesh Mahadasa (MSc), *Incremental Graph Mining.*
- Jeremy Schwartz (undergraduate), *Graph Pattern Generation.*
- Hao Henry Fang (undergraduate), *Pattern-Aware Graph Mining on Weighted Graphs.*
- Daniel Gomes Maia Filho (undergraduate), *Workload-Balancing in Incremental Graph Mining.*
- Richard Dong (undergraduate), *Parallel Frequent Subgraph Mining.*

### Reviewing for Journals & Conferences

SOSP | ASPLOS | OSDI | VLDB | EuroSys | ATC | TODS | PACT | ICS | ICDCS.

### President of the Computing Science Student Society

- Taught undergraduate workshops on git and Linux software development.
- Organized week-long student trip to Silicon Valley for tours and networking events.
- Ran Mountain Madness hackathon rewarding innovative and ingenious student projects.

### Technical Writer at BC Children's Society

- Drafted and edited program and funding proposals to the Ministry of Children and Families for new initiatives to assist children and youth with support needs.
- Revised internal training and reference manuals.

## Honours & Awards

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| Best Poster Award - Anti-Vertex For Neighborhood Constraints | 2022 |
| SFU Computing Science Graduate Fellowship | 2019, 2021, 2022, 2023 |
| Clark Wilson LLP Graduate Scholarship | 2022 |
| Best Poster Award - Peregrine: A Pattern-Aware Graph Mining System | 2020 |
| SFU Vice President-Research Undergraduate Student Research Award | 2018 |
| Gordon M. Shrum Major Entrance Scholarship | 2014 |